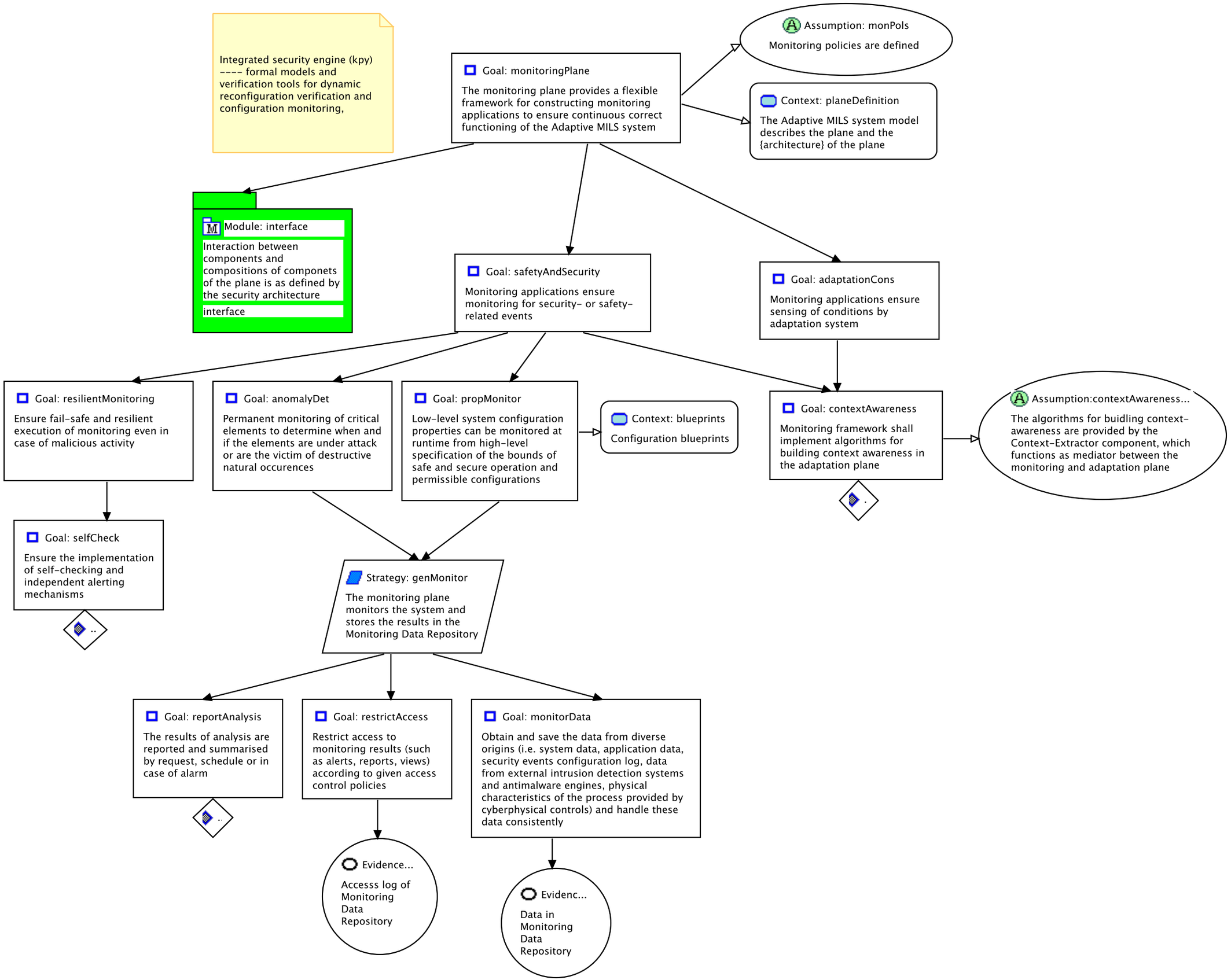
Integrated security engine (kpy) ---- formal models and verification tools for dynamic reconfiguration verification and configuration monitoring,
Goal: monitoringPlane
The monitoring plane provides a flexible framework for constructing monitoring applications to ensure continuous correct functioning of the Adaptive MILS system
Assumption: monPols
Monitoring policies are defined
Context: planeDefinition
The Adaptive MILS system model describes the plane and the {architecture} of the plane
Module: interface
Interaction between components and compositions of componets of the plane is as defined by the security architecture
interface
Goal: safetyAndSecurity
Monitoring applications ensure monitoring for security- or safety-related events
Goal: adaptationCons
Monitoring applications ensure sensing of conditions by adaptation system
Goal: resilientMonitoring
Ensure fail-safe and resilient execution of monitoring even in case of malicious activity
Goal: anomalyDet
Permanent monitoring of critical elements to determine when and if the elements are under attack or are the victim of destructive natural occurences
Goal: propMonitor
Low-level system configuration properties can be monitored at runtime from high-level specification of the bounds of safe and secure operation and permissible configurations
Context: blueprints
Configuration blueprints
Goal: contextAwareness
Monitoring framework shall implement algorithms for building context awareness in the adaptation plane
Assumption:contextAwareness...
The algorithms for buidling context-awareness are provided by the Context-Extractor component, which functions as mediator between the monitoring and adaptation plane
Goal: selfCheck
Ensure the implementation of self-checking and independent alerting mechanisms
Strategy: genMonitor
The monitoring plane monitors the system and stores the results in the Monitoring Data Repository
Goal: reportAnalysis
The results of analysis are reported and summarised by request, schedule or in case of alarm
Goal: restrictAccess
Restrict access to monitoring results (such as alerts, reports, views) according to given access control policies
Goal: monitorData
Obtain and save the data from diverse origins (i.e. system data, application data, security events configuration log, data from external intrusion detection systems and antimalware engines, physical characteristics of the process provided by cyberphysical controls) and handle these data consistently
Evidence...
Accesss log of Monitoring Data Repository
Evidenc...
Data in Monitoring Data Repository